

A Soft Introduction to CMA-ES Technique

Optimization Algorithm Team

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Introduction

Optimization with Gradient Descents

Random Search

Evolution Strategy: (μ, λ) -ES

Ill-Conditioned Quadratic

Introduction

The objective of the study is to understand the progression

random search → evolution strategy → adaptive Gaussian search → CMA-ES

After: we shall see how important it is to adapt a full covariance matrix can make the algorithm nearly insensitive to rotations and anisotropic scalings of the search landscape.

Optimization with Gradient Descents

Let $D \subset \mathbb{R}^d$ and $f : D \rightarrow \mathbb{R}$. We want to minimize the function f

$$\min_{x \in D} f(x) \tag{1}$$

If f is smooth (i.e., continuously differentiable) then to solve (1), we can use gradient descent methods which generate the solution by updating

$$x_{n+1} = x_n - \sigma \nabla f(x_n)$$

Search information

The optimization algorithm uses gradient to go to the descent directions.

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Challenge: f is not differentiable!



Derivative-Free Optimization

We are here to answer three questions:

1. What information does a gradient use that a population-based methods does not have directly?

Ans Gradient methods use the first-order derivative of the objective function. The gradient provides the descent direction in the search process to obtain the minimizer which the population-based methods do not access. Derivative-free methods evaluate the objective function values at the a set of candidate points and select the best that improves the function value. These methods however, help us to do optimization with nonsmooth functions which is practical in real-world applications.

2. Why can derivative-free methods still recover preferred directions of descent?

Ans Yes, DFO can recover the descent direction even without gradient information. To obtain so, DFO implicitly approximate the gradient-like information using the fitness values (function values) and they structure their search so that at least one trial direction is guaranteed to be descent direction.

3. What difficulties arise when d is large?

Random Search

If f is differentiable, we reach the optimizer based on the gradient information, now we don't have this information any more. How can we reach the optimizer?

Random Search

Having a starting point $m_0 \Rightarrow$ generate random points around $m_0 \Rightarrow$ calculate the function values f

This is as stated " Finding a needle in an n-dimensional haystack" [1, 2].

Search information

The optimization algorithm uses a set of randomly generated points based on probability distribution.

Evolution Strategy: (μ, λ) -ES

Algorithm 1: $(1, \lambda)$ -ES Iteration: Isotropic Search

Input: A point $m_0 \in \mathbb{R}^d$, step size $\sigma > 0$, population size λ

```
1 for  $k \leftarrow 1$  to  $n$  do
2   for  $i \leftarrow 1$  to  $\lambda$  do
3     Sample  $Z_i \sim \mathcal{N}(0, I_d)$ 
4     Compute  $X_i \leftarrow m_k + \sigma Z_i$  //  $X_i \sim \mathcal{N}(m_0, \sigma^2 I_d)$ 
5   Compute fitness value  $f(X_1), \dots, f(X_\lambda)$ 
6    $i^* \leftarrow \arg \min_{i \in \{1, \dots, \lambda\}} f(X_i)$ 
7    $m_{k+1} \leftarrow X_{i^*}$ 
8 return  $m_n$ 
```

Algorithm 2: (μ, λ) -ES Iteration: Isotropic Search

Input: A point $m_0 \in \mathbb{R}^d$, step size $\sigma > 0$, population size λ , number of selected parent $\mu \leq \lambda$, weighted factor ω_i s.t. $\sum_{i=1}^{\mu} \omega_i = 1$.

1 **for** $k \leftarrow 1$ **to** n **do**

2 **for** $i \leftarrow 1$ **to** λ **do**

3 $Z_i \sim \mathcal{N}(0, I_d)$

4 $X_i = m_k + \sigma Z_i$

5 Compute $f(X_i)$ and sort offspring by fitness:

$$X_{1:\lambda}, X_{2:\lambda}, \dots, X_{\lambda:\lambda},$$

where $X_{1:\lambda}$ is the best, $X_{2:\lambda}$ is the second-best, and so on.

6 Choose μ best candidates and recombine

$$m_{n+1} = \sum_{i=1}^{\mu} \omega_i X_{i:\lambda}.$$

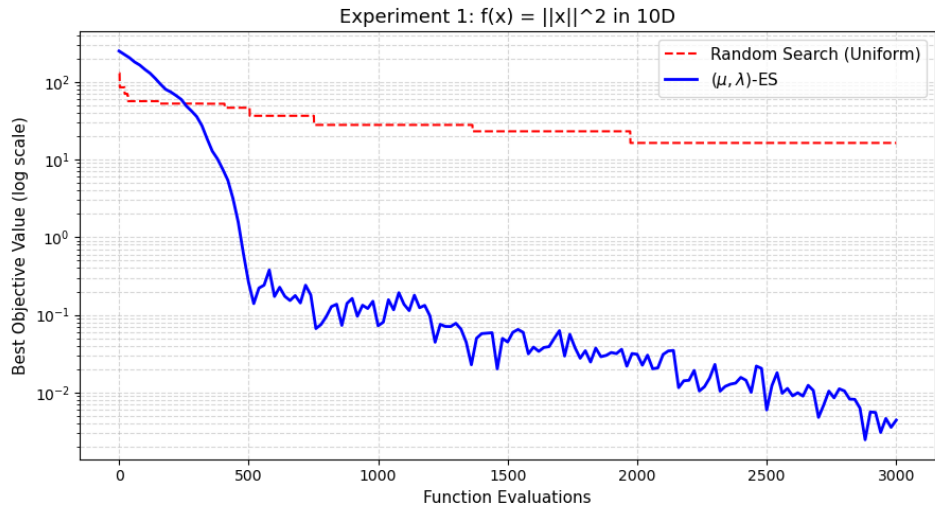


Figure 1: Comparison between random search with uniform distribution and (μ, λ) -ES.

III-Conditioned Quadratic

Anisotropic level set

- With covariance adaptation, we see that to obtain the solution it goes through to the optimal solution.
- While the isotropic search cannot. It more or less is taking a detour to obtain the solution.
- A central observation is that isotropic mutation is inefficient when the successful search from a thin elongated ellipsoid.

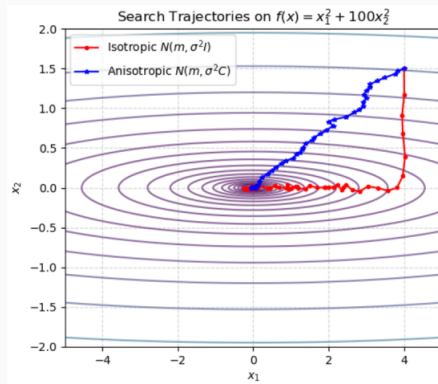


Figure 2: Function f has anisotropic level set. (Steep valley in x_2 and flat in x_1).

Updated Search Information

The optimization algorithm uses λ randomly generated points called *offsprings* and

Anisotropic search

In stead of sampling from

$$X \sim \mathcal{N}(m, \sigma^2 I),$$

we consider the offspring of the form

$$X \sim \mathcal{N}(m, \sigma^2 C)$$

where C is symmetric positive definite.

What does the covariance matrix C do here?

A picture here similar to the one that prof explained in the first day.

The picture equals =

- The matrix C determines the shape of and orientation of the search distribution.
- The idea of covariance is relatively simple: Directions in which successful stels have repeated ly occurred should receive larger variance in the future generations.

Search directions

The optimization algorithm learns not only where to search thought the starting point m but also how to to search through C .

References

- [1] Granino Arthur Korn. *Random-Process Simulation and Measurements*. McGraw-Hill, New York, 1966.
- [2] R.C. White. *A survey of random methods for parameter optimization*. EUT report. E, Fac. of Electrical Engineering. Technische Hogeschool Eindhoven, 1970.

**Thank you for your attentions!!!
:-)**